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Partial Solutions to "Universal" Problems: A Fundamental Introduction

Master's Thesis

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Master of Science in Engineering (MSc)

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Hagenberg

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Affidavit

I hereby declare that I have written the presented thesis independently and have used only the references and aids indicated. All passages taken verbatim or paraphrased from the references are clearly identified as such. This thesis has not been submitted, in whole or in part, in the same or a similar form, to any other examination authority, and is identical to the electronically submitted text document.

Hagenberg, July 1, 2027

Alex A. Wiseguy

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Preface

Abstract

This should be a 1-page (maximum) summary of your work in English.

Kurzfassung

An dieser Stelle steht eine Zusammenfassung der Arbeit, Umfang max. 1 Seite. ...

Chapter 1

Introduction

Chapter 2

Writing a Thesis

Chapter 3

Working with LaTeX

Chapter 4

Figures, Tables, Source Code

Chapter 5

Mathematical Elements, Equations and Algorithms

Chapter 6

Using Literature and other Resources

[1]

Chapter 7

Printing the Manuscript

Chapter 8

Closing Remarks

Appendix A

Technical Details

Appendix B

Supplementary Materials

List of supplementary data submitted to the degree-granting institution for archival storage (in ZIP format).

B.1 PDF Files

Path: /

thesis.pdf . . . . . Master/Bachelor thesis (complete document)

B.2 Media Files

Path: /media

*.ai, *.pdf . . . . . Adobe Illustrator files
*.jpg, *.png . . . . . raster images
*.mp3 . . . . . audio files
*.mp4 . . . . . video files

B.3 Online Sources (PDF Captures)

Path: /online-sources

Reliquienschrein-Wikipedia.pdf [2]

Appendix C

Questionnaire

Appendix D

LaTeX Source Code

Documentation of Generative AI Tool Usage

Responsible use of the AI tools

Statements on responsible use of AI tools	**Confirmation: Yes**
I/we confirm that the outputs generated by AI tools have been verified for accuracy and corrected where required.	☒
I/we acknowledge that the responsibility for the thesis lies with the author(s), and not with the AI tools or any other person.	☒

Detailed activities for which the tools were used

Activity	**Description**	**Tools used (if any)**	**Description of the tool usage** (type of usage, affected sections of the paper/thesis, etc.)
Idea generation & conceptualization	• Generating ideas, objectives and research questions • Identifying and defining relevant concepts and theoretical models • Development of a conceptual framework for the content	—	—
Literature search and analysis	• Search for relevant literature • Reviewing potentially relevant literature • Summaries of relevant literature	Perplexity	Search for literature on ... (Chapter 2)
Methodology	• Search for an appropriate methodology • Development and adaptation of the methodology to the research question(s)	—	—

Activity	**Description**	**Tools used (if any)**	**Description of the tool usage** (type of usage, affected sections of the paper/thesis, etc.)
Coding	• Creation and documentation of code, algorithms, and software • Testing and debugging of existing code, algorithms, software • Understanding and analysis of existing code, algorithms, and software	—	—
Data collection and analysis	• Collection of primary or secondary data • Qualitative data analysis (including summarizing and coding) • Quantitative data analysis (including statistics) • AI-assisted mathematical, computational, or other formal techniques for modeling, simulation, and analytics	—	—
Interpretation and Validation	• Interpretation of results • Derivation of implications for research and practice • Verification of the overall replicability/reproducibility of the results and other research outputs	—	—
Structuring and planning the text	• Structure of the thesis • Structuring of individual sections of the thesis (e.g., bulletpoint outlines per section)	—	—
Generating the text	• Generation of text for topics in various sections of the thesis (including title and abstract)	—	—
Translating text	• Translation of text written by the author(s) • Translation of text written by others	—	—
Reviewing & editing the text	• Critical review, feedback or revision of the content or grammar of the thesis • Proofreading • Rephrasing or paraphrasing text • Shortening / expanding the text	DeepL Write	Proofreading of Chapters 1–5
Citation Management	• Creation of the reference list • Formatting of the references	—	—

Further use of AI tools:

—

Source reference: Gimpel, H., Dilger, P., Lämmermann, L., Urbach, N. (2023). COGNISANCE: Declaration of Generative AI Tool Usage in Higher Education Tests. Research Center for Information Management, May 22, 2023. https://doi.org/10.17605/OSF.IO/VGXF7. Adapted by the University of Applied Sciences Upper Austria. Licensed under CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0/).

References

Literature

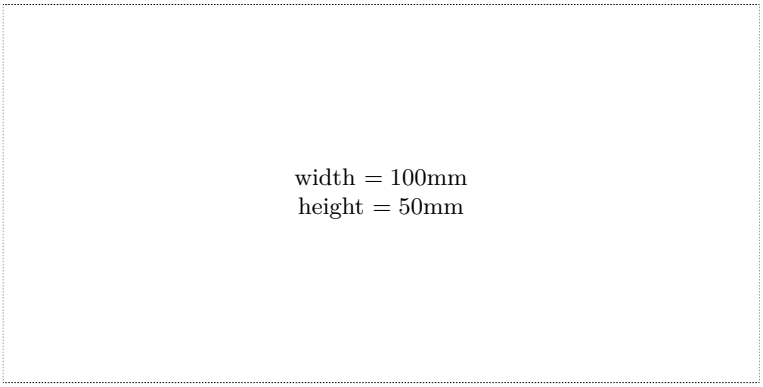
[1] Hubert M. Drake, Milton D. McLaughlin, and Harold R. Goodman. *Results obtained during accelerated transonic tests of the Bell XS-1 airplane in flights to a MACH number of 0.92*. Tech. rep. NACA-RM-L8A05A. Edwards, CA: NASA Dryden Flight Research Center, Apr. 19, 1948. URL: https://ntrs.nasa.gov/api/citations/19930085320/downloads/19930085320.pdf (cit. on p. 6).

Online sources

[2] *Reliquienschrein*. June 10, 2026. URL: https://de.wikipedia.org/wiki/Reliquienschrein (visited on 09/13/2026).

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